GELDIUM

Customer Delinquency Exploratory Data Analysis

Data quality, financial behavior, segment risk and statistical evidence

Tata GenAI-Powered Data Analytics | Portfolio case study

500 customer records | 19 original fields | 29 final fields | 16% delinquency rate

Prepared from the supplied 24-page Geldium EDA report. All numerical findings are reproduced from that report; no additional hypothesis tests or predictive model results are claimed.

Executive summary

This exploratory study assessed 500 customer accounts to characterize delinquency prevalence, evaluate data quality, and identify candidate indicators for subsequent credit-risk modeling. The observed delinquency rate was 16.0% (80/500), creating a moderately imbalanced classification problem.

KEY MEASURE	RESULT	IMPLICATION
Accounts analyzed	500	Modest sample; segment estimates are uncertain
Delinquent accounts	80 (16.0%)	Accuracy alone is misleading
Missing data before cleaning	70 cells across 3 fields	Imputation needed; preserve relevant flags
Final dataset	500 × 29; no missing values	Ready for careful modeling preparation
Numerical correlations	All |r| < 0.05	No strong standalone linear signal
Categorical tests	All reported p-values > 0.05	No confirmed standalone association

Principal conclusions

1. Segment rates vary, but the report does not establish reliable predictors. Business cardholders (21.30%), unemployed customers (19.35%), and customers with missing loan balances (24.14%) have higher observed rates; uncertainty and small subgroups prevent firm conclusions.

2. Six-month payment behavior is unexpectedly weakly related to the target in this sample. Confirm how the target and payment-history fields were defined and timed before modeling.

3. Four credit-utilization values exceed 100%; they may be legitimate but warrant validation against source definitions.

4. The recommended next step is a leakage-safe, stratified modeling workflow with PR-AUC, recall, calibration and fairness checks, rather than operational credit decisions based on these descriptive differences.

Report map

01 Data and quality • 02 Distribution patterns • 03 Delinquency and segments • 04 Payment behavior • 05 Statistical evidence • 06 Recommendations and limitations

01 Dataset and data quality

Study scope

The source report describes 500 customers and 19 original variables spanning age, income, credit score, credit utilization, missed payments, loan balance, debt-to-income ratio, employment, account tenure, card type, location, six monthly payment statuses and the binary delinquency outcome. The final analysis dataset has 29 fields after cleaning and feature engineering.

QUALITY CHECK	FINDING
Unique customers	500; no duplicate Customer_ID values

QUALITY CHECK	FINDING
Missing Income	39 of 500 (7.8%)
Missing Loan Balance	29 of 500 (5.8%)
Missing Credit Score	2 of 500 (0.4%)
Total missing cells	70 before imputation; 0 after
Payment-history consistency	All 500 records satisfy On-time + Late + Missed = 6
Utilization above 100%	4 records (0.8%); retained for review

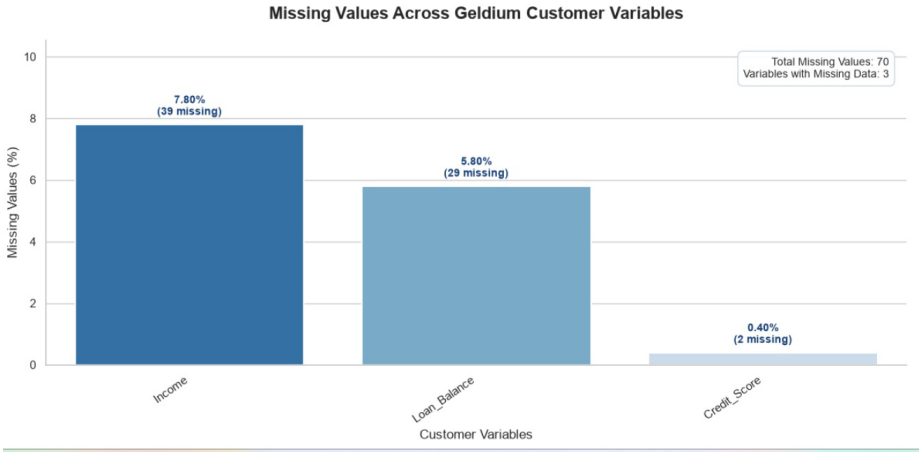


Figure 1. Missingness before cleaning. Original source figure.

Cleaning approach

The original report states that mean imputation was applied to Income, Loan_Balance and Credit_Score; employment labels were standardized; and Loan_Balance_Missing was retained as a binary indicator. For future prediction, imputation parameters must be estimated from training folds only. The full 19-to-29 field derivation should be documented in the accompanying notebook.

02 Distribution patterns

The numerical variables cover diverse customer profiles. Means and medians are generally similar, while income, loan balance and credit scores exhibit broad dispersion. The source report states that no extreme observations were detected using the 1.5 × IQR rule; this does not supersede domain-specific validation.

VARIABLE	CENTER	NOTABLE OBSERVATION
Age	Mean 46.27; median 46.5	Broad age distribution
Income	Mean $108,379.89	Q1 $62,295; Q3 $155,734
Credit Score	Mean 577.72	Q1 418; Q3 727
Credit Utilization	Mean 49.1%	4 observations above 100%
Loan Balance	Mean $48,654.43	Median $45,776
Debt-to-Income	Mean approximately 0.30	Concentrated near 30%

VARIABLE	CENTER	NOTABLE OBSERVATION
Missed Payments	Mean 2.97	Discrete 0–6
Account Tenure	Mean 9.74	Observed range 0–19

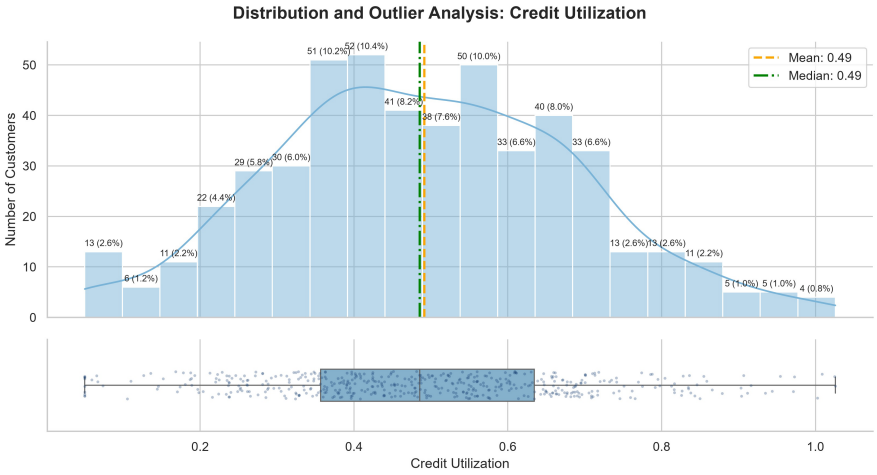


Figure 2. Credit utilization distribution, including mean, median and customer-level spread. Original source figure.

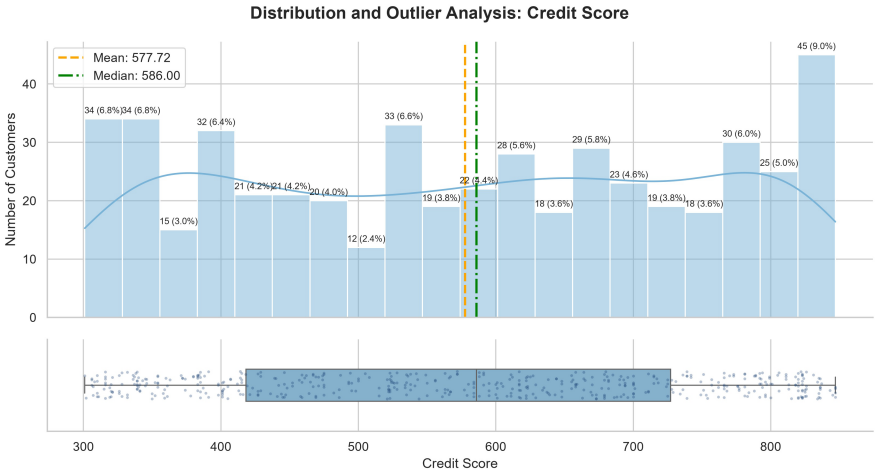


Figure 3. Credit score distribution and boxplot. Original source figure.

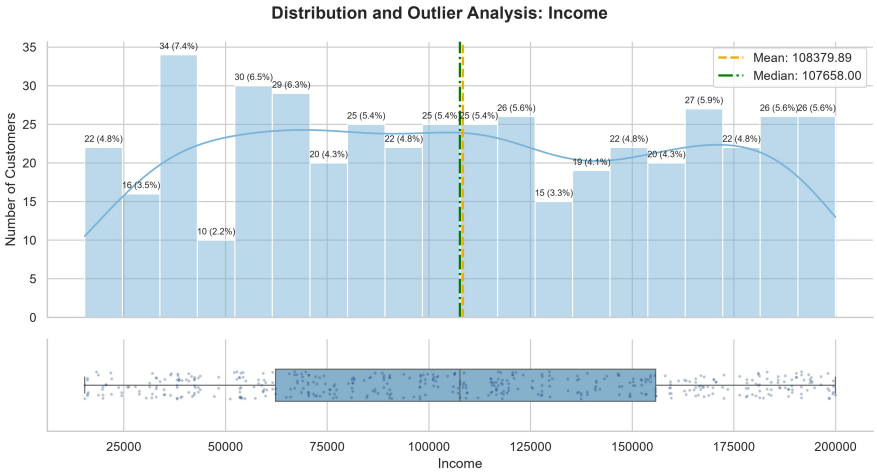


Figure 4. Income distribution and boxplot. Original source figure.

Distribution interpretation

The broad spreads in income and credit score suggest heterogeneous customer profiles, but dispersion by itself does not imply predictive power. The four utilization observations above 100% could reflect over-limit balances or measurement conventions; the report does not provide enough evidence to determine which.

03 Delinquency prevalence and segment comparisons

Of 500 customers, 420 were non-delinquent and 80 delinquent. A model predicting only the majority class would achieve 84% accuracy but zero recall for delinquency; therefore, accuracy alone is inadequate.

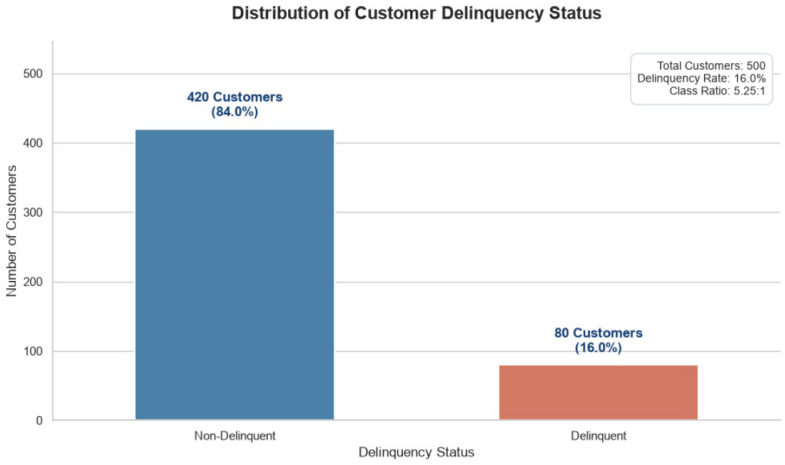


Figure 5. Class distribution: 420 non-delinquent (84%) versus 80 delinquent (16%). Original source figure.

Observed categorical differences

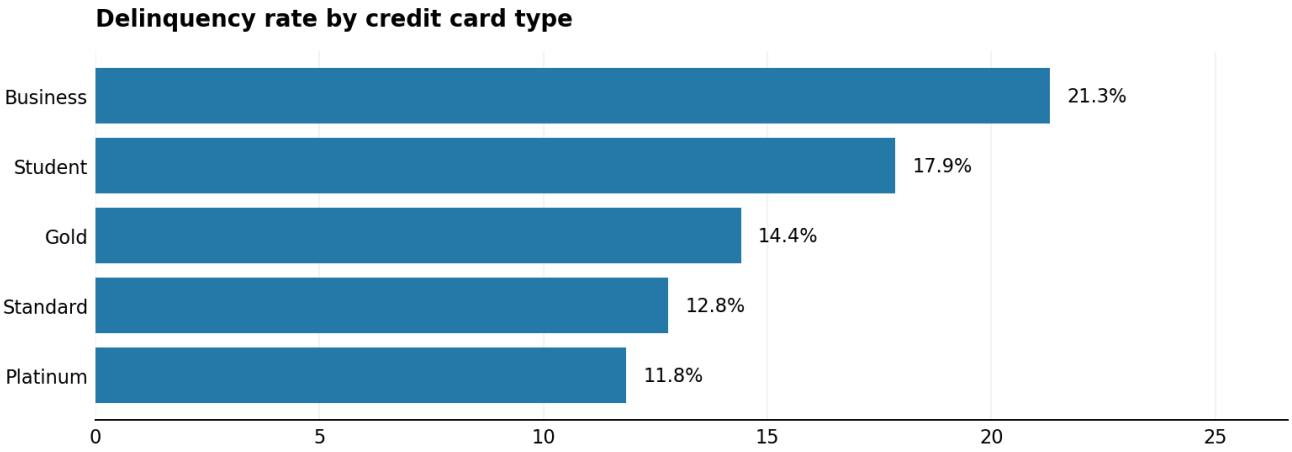


Figure 6. Observed rates by card type. Descriptive only; no statistically significant overall association was established.

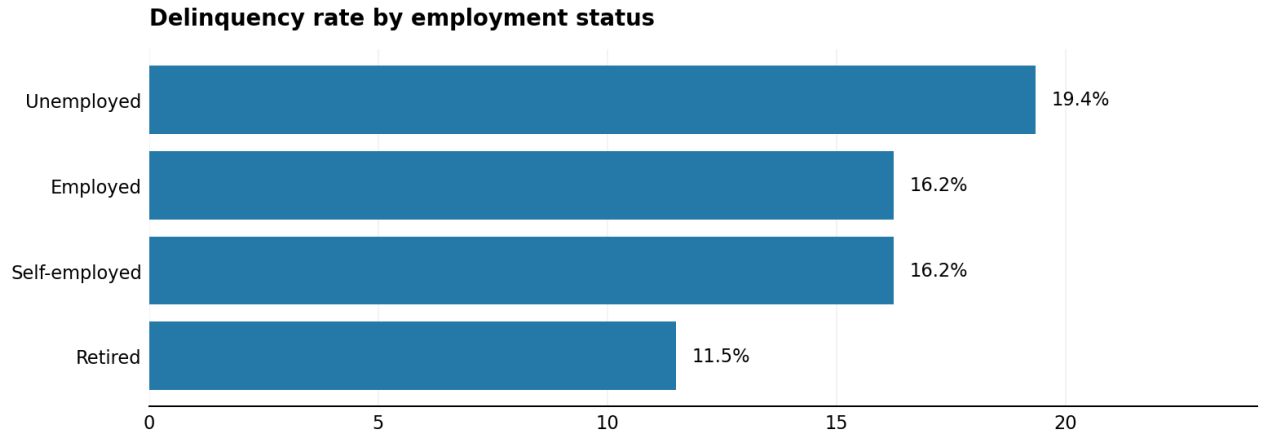


Figure 7. Observed rates by employment status. These differences are not causal evidence.

SEGMENT	N	DELINQUENT	RATE
Business card	108	23	21.30%
Platinum card	76	9	11.84%
Unemployed	93	18	19.35%
Retired	87	10	11.49%
Los Angeles	107	21	19.63%
New York	92	11	11.96%

Numerical risk bands

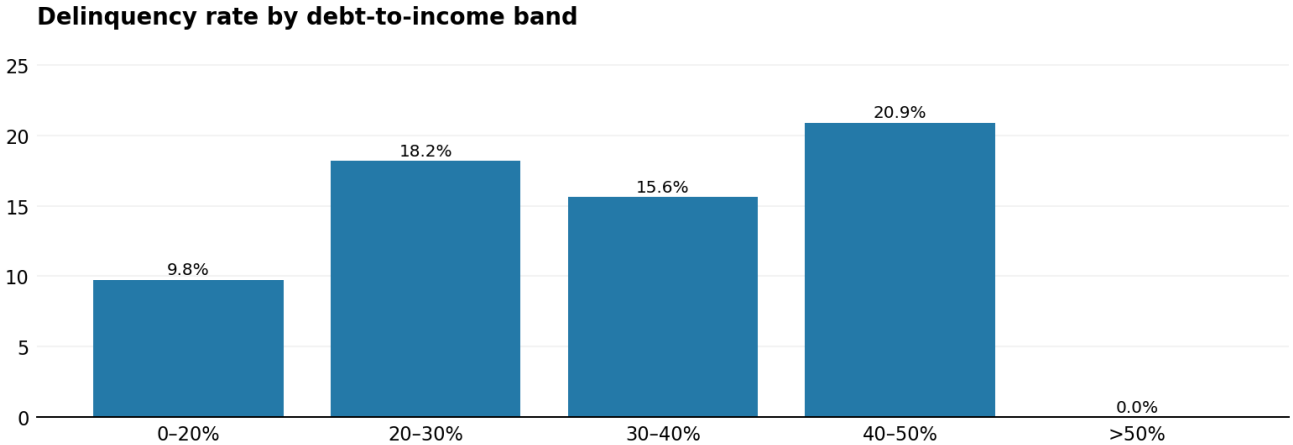


Figure 8. Rates are non-monotonic. The >50% band contains only seven customers and zero delinquency events.

CREDIT UTILIZATION	N	EVENTS	RATE
0–30%	83	10	12.05%
30–50%	182	31	17.03%
50–70%	166	28	16.87%

CREDIT UTILIZATION	N	EVENTS	RATE
70–100%	65	11	16.92%
Above 100%	4	0	0.00%

Credit-score delinquency rates were likewise non-monotonic: 14.29% at 300–579, 16.67% at 580–669, 17.46% at 670–739, 12.96% at 740–799 and 23.33% at 800–850. The high-score result is counterintuitive and should be checked for sampling variation and target-definition effects before interpretation.

04 Six-month payment behavior

Each of Month_1 through Month_6 is categorical: On-time, Late or Missed. The report engineers OnTime_Months_6M, Late_Months_6M, Missed_Months_6M and Payment_Severity_6M, assigning scores 0, 1 and 2 respectively.

MONTH	ON-TIME	LATE	MISSED
Month 1	177	159	164
Month 2	160	173	167
Month 3	169	169	162
Month 4	159	181	160
Month 5	162	151	187
Month 6	160	172	168

Across delinquency classes, mean Late_Months_6M was 2.02 versus 1.95, mean Missed_Months_6M was 2.02 versus 2.01, and mean Payment_Severity_6M was 6.05 versus 5.98 (non-delinquent versus delinquent). All four engineered behavioral features had reported absolute target correlations below 0.025.

Analytical implication: The apparent weakness of payment behavior as a predictor deserves scrutiny. Verify whether the outcome was generated independently of the six-month statuses, whether the feature window precedes the target, and whether any fields leak future information. The supplied report does not establish these points.

05 Statistical evidence and uncertainty

Numerical association

VARIABLE	PEARSON r
Income	+0.0439
Account Tenure	-0.0398
Credit Score	+0.0348
Debt-to-Income Ratio	+0.0344
Credit Utilization	+0.0342
Missed Payments	-0.0265
Age	+0.0225
Loan Balance	-0.0041

All absolute Pearson correlations are below 0.05. This supports a conclusion of negligible observed standalone linear association, not proof that the variables contain no nonlinear or interaction information.

Categorical association

FEATURE	CRAMÉR'S V	p-VALUE
Month_1	0.0948	0.1058
Credit_Card_Type	0.0938	0.3544
Month_4	0.0914	0.1236
Location	0.0692	0.6637
Month_5	0.0651	0.3469
Employment_Status	0.0649	0.5503
Month_6	0.0543	0.4778
Month_2	0.0287	0.8137
Month_3	0.0276	0.8261

None of the nine reported Chi-square tests reaches $p < 0.05$; all reported Cramér's V values are below 0.10. Before relying on these tests, expected cell counts should be checked. Multiple exploratory comparisons also increase the chance of misleading extremes.

Loan Balance missingness

STATUS	NON-DELINQUENT	DELINQUENT	RATE
Available (n=471)	398	73	15.50%
Missing (n=29)	22	7	24.14%

The source report gives Fisher’s exact odds ratio = 1.7347, p = 0.2920. This is an exploratory difference, not a statistically confirmed association. A confidence interval would better communicate its uncertainty.

06 Business implications and recommended next steps

PRIORITY	RECOMMENDED ACTION	REASON
1	Verify outcome definition and feature timing	Payment-history patterns are unexpectedly weak; avoid leakage
2	Implement train-only preprocessing	Prevent optimistic validation from imputation leakage
3	Benchmark simple and nonlinear models	Test whether interactions add value
4	Use stratified validation and PR-AUC	Only 16% of records are positive
5	Report uncertainty and calibration	Segment rates alone are not decision rules
6	Assess fairness and data suitability	Location and other proxies can pose credit-decision risks

Suggested model evaluation protocol

Begin with a majority-class or simple probability baseline and logistic regression; compare with Random Forest and Gradient Boosting using stratified cross-validation. Evaluate recall, precision, F1, ROC-AUC, PR-AUC, calibration and confusion matrices. Keep a held-out test set untouched until final assessment. The present report does not include trained-model results.

Limitations

The dataset contains only 500 records and 80 events; some bands have fewer than 10 customers. Segment comparisons are exploratory; confidence intervals were not reported. The data provenance, variable definitions, target observation window, currency assumptions and complete engineering lineage are not documented in the supplied PDF. Mean imputation was performed in the EDA, but training-time preprocessing must be fitted separately. No out-of-sample predictive performance is available.

Final conclusion

The Geldium EDA provides a sound descriptive foundation: it characterizes 500 customer accounts, documents 16% delinquency prevalence, addresses missingness, engineers six-month behavioral features and evaluates numerical and categorical associations. The analysis consistently finds weak standalone associations, so the observed segment differences should not be used as validated risk rules.

Decision: Advance to carefully validated predictive modeling only after confirming data provenance, target timing and preprocessing controls. The main value of this

analysis is disciplined data understanding, not an established credit-risk scoring system.

Source and scope note

Prepared exclusively from the user-supplied Geldium Customer Delinquency Analysis - Comprehensive Exploratory Data Analysis (24-page PDF). Original distribution figures are reused from that PDF; risk-focused bar charts are newly rendered from its reported numbers. No new statistical calculations, external validation, or model experiments were performed.